SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

QUESTIONS: **Total Marks:25**

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Ans) A Convolutional Neural Network (CNN) is more suitable for image classification than a traditional Fully Connected Neural Network (FCNN) because CNNs are specifically designed to process data with spatial relationships, such as images. In a fully connected network, the entire image is usually converted into a one-dimensional vector, and every neuron is connected to every neuron in the next layer. This results in a very large number of parameters and high computational and memory requirements, especially for high-resolution X-ray images. In contrast, a CNN uses convolutional layers that apply small learnable filters to local regions of the image. These filters automatically detect important features such as edges, corners, textures, shapes, and anatomical structures. Pooling layers reduce the spatial size of feature maps, which decreases computation and makes the network more robust to small changes in object position. As the network becomes deeper, it learns increasingly complex features from simple edges to meaningful patterns. CNNs also use parameter sharing, meaning the same filter weights are reused across different locations in the image, significantly reducing the number of parameters. Therefore, CNNs preserve spatial information, require fewer parameters, extract useful visual features automatically, and provide better performance for image classification. For medical X-ray analysis, CNNs are therefore more effective than traditional fully connected networks.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Ans) In a CNN-based industrial defect detection system, different layers perform different functions to identify defects effectively. The convolutional layer is mainly responsible for feature extraction by applying filters to the input image. These filters learn important patterns such as edges, lines, textures, cracks, and surface irregularities. The activation layer, commonly using the ReLU function, introduces non-linearity into the network, allowing it to learn complex patterns that cannot be learned using only linear operations. Pooling layers reduce the spatial dimensions of feature maps while retaining the most important information, thereby reducing computational requirements and making the system more efficient. Finally, fully connected layers combine the extracted high-level features and perform the final classification, such as deciding whether a product is defective or defect-free. The filters in early convolutional layers usually learn simple features such as edges, corners, and basic textures. Deeper layers combine these simple features to learn complex patterns such as cracks, scratches, holes, abnormal shapes, and other manufacturing defects. Thus, convolution, activation, pooling, and fully connected layers work together to gradually transform a raw product image into meaningful information and improve the accuracy of defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for largescale image-processing applications.

Ans) Parameter sharing is one of the most important advantages of CNNs compared with fully connected networks. In a fully connected network, each connection has its own independent weight, so processing a high-resolution image requires a very large number of parameters. This increases memory usage, computational cost, training time, and the possibility of overfitting. In CNNs, the same convolutional filter weights are reused across different spatial locations of the image. For example, a single edge-detection filter can detect the same type of edge whether it appears on the left, right, top, or bottom of an image. This greatly reduces the number of parameters and allows the network to learn efficiently from large datasets. Parameter sharing also provides a degree of translation equivariance because the same feature can be detected at different locations. Pooling and other CNN operations can further provide some translation tolerance. For a smart surveillance system processing thousands of high-resolution images, parameter sharing significantly reduces memory requirements and computational complexity while allowing the network to detect common visual features efficiently. Therefore, parameter sharing makes CNNs highly suitable for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Ans) AlexNet and ResNet are important CNN architectures, but they differ significantly in depth and design. AlexNet was introduced as a relatively deep CNN with about eight learned layers and uses convolutional layers followed by fully connected layers for classification. It uses convolutional filters to learn visual features and played an important role in demonstrating the effectiveness of deep learning for image recognition. However, as networks become deeper, training becomes difficult because of problems such as vanishing gradients and degradation of performance. ResNet, or Residual Network, addresses these problems by introducing residual or skip connections. Instead of forcing every layer to learn a complete transformation, residual blocks allow information and gradients to pass directly through shortcut connections. This makes it possible to train much deeper networks, such as ResNet-50, ResNet-101, and ResNet152. ResNet generally has better feature extraction capability because deeper layers can learn more complex and high-level visual representations. Although deeper ResNet models can require considerable computational resources, their improved architecture makes training more stable and reduces the vanishing-gradient and degradation problems. For a complex image recognition application, ResNet is generally the better choice because it supports very deep networks, extracts more sophisticated features, and usually provides better classification performance than AlexNet.

Q5. Regularized CNN vs Non-Regularized CNN

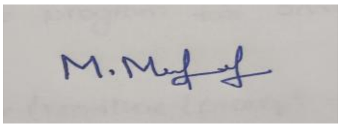
An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Ans) A CNN without regularization may achieve very high training accuracy but perform poorly on unseen images because the model can memorize the training data instead of learning general patterns. This problem is called overfitting. A regularized CNN uses techniques such as dropout, data augmentation, and batch normalization to improve generalization. Dropout randomly deactivates some neurons during training, preventing the network from depending too heavily on particular features and reducing overfitting. Data augmentation artificially increases the size and diversity of the training dataset by applying transformations such as rotation, flipping, cropping, scaling, and brightness changes to leaf images. This is particularly useful for plant disease classification because leaves may appear in different orientations, sizes, lighting conditions, and backgrounds. Batch normalization normalizes intermediate activations during training, which can improve training stability and often allows the network to train faster. These techniques may add some computational overhead, but the improvement in generalization is usually more important for real-world applications. Therefore, for plant disease classification, a combination of **data augmentation, dropout, and batch normalization** is recommended. Data augmentation is especially important because it provides more diverse examples, while dropout helps prevent overfitting and batch normalization improves training stability. Together, these techniques can produce a more reliable CNN that performs well not only on training images but also on unseen real-world leaf images.

Code of Conduct:

I M.MEGHANADH(192325026)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies

Comparison & Differentiation	**25%**	**Clearly compares multiple aspects (methods, results, limitations) with strong differentiation**	**Good comparison with minor gaps**	**Basic comparison with limited depth**	**No clear comparison; mostly descriptive**
Critical Evaluation	**25%**	**Critically evaluates strengths, weaknesses, and implications with strong justification**	**Provides evaluation with some justification**	**Limited evaluation; lacks depth**	**No critical evaluation provided**
Synthesis & Insight Generation	**20%**	**Generates meaningful insights and conclusions from comparisons**	**Some insights derived from comparison**	**Limited or unclear insights**	**No insights generated**
Clarity	**10%**	**Information is clearly presented (tables/figures) with logical structure**	**Generally clear with minor issues**	**Presentation lacks clarity or structure**	**Poor presentation and difficult to understand**